ShelfHelp: Vision Document

Version: 1.0

1. Introduction

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The purpose of this document is to collect, analyze, and define high-level needs and features of the ShelfHelp Smart Library Management System. It focuses on the capabilities needed by the stakeholders and the target users, and why these needs exist.

1.1 References

- Project Assignment 1 (PA1-2026): Project Proposal.
- Project Assignment 2 (PA2-2026): Guidelines and Requirements.

2. Positioning

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2.1 Problem Statement

The problem of	inefficient, manual university library management using spreadsheets, paper cards, or outdated systems
affects	university students and librarians
the impact of which is	students struggling to discover relevant reading materials based on research topics and missing due dates, while librarians waste significant time on manual record-keeping, chasing overdue books, and compiling reports.
a successful solution would be	digitizing the whole lending cycle with automated reminders, real-time inventory tracking, and integrating an AI research assistant to help students find books using natural language.

2.2 Product Position Statement

For	university students and librarians
Who	need an efficient, automated, and smart way to borrow, discover, and manage library resources
The (product name)	ShelfHelp
That	provides a comprehensive responsive web application that digitizes the lending cycle, introduces a sustainable membership model, and offers AI-powered book

	recommendations
Unlike	traditional paper-based workflows, spreadsheets, or legacy library catalog systems
Our product	bridges the gap between digital requests and physical counter handovers, automates fee/fine calculations, and features a Retrieval-Augmented Generation (RAG) chatbot that recommends books actually held by the library based on natural-language topics.

3. Stakeholder and User Descriptions

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3.1 Stakeholder Summary

Name	Description	Responsibilities
University Institution / Board	The governing body of the university providing the budget and oversight.	Ensures the library operates efficiently; monitors lost assets and overall revenue from memberships/fines; requires accurate statistical reporting.
IT Department	The technical staff of the university.	Ensures system uptime, security, and maintenance of the web application and database environment.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Student (Reader)	University students with a university-issued email and student ID.	Searches for books, requests/returns loans, manages memberships, pays fees/fines, writes reviews, and interacts with the AI chatbot.	University Institution
Librarian (Admin)	Library staff managing daily operations at the library counter and shelves.	Manages catalog/inventory, hands over and receives books, records physical book conditions, processes in-person payments, plans acquisitions, and generates reports.	University Institution

3.3 User Environment

- **Platform:** Responsive web application accessible on Windows, macOS, Android, and iOS.
- **Usage Context:**
  - **Students:** Designed mobile-first. They mostly use smartphones between classes (e.g., scanning a book's barcode at a shelf, checking due dates, chatting with AI) and laptops when studying or writing a thesis.
  - **Librarians:** Designed desktop-first. They mainly work on desktop computers at the library counter, with occasional tablet or phone use on the shelves for barcode scanning and inventory checks.
- **Hardware Interfacing:** The device's camera is utilized for avatar capture and scanning physical book barcodes.

3.4 Summary of Key Stakeholder or User Needs

Need	Priority	Concerns	Current Solution	Proposed Solutions
Semantic Discovery	High	Keyword search fails when students only know a research topic, not specific titles.	Wandering physical shelves or using inaccurate keyword catalogs.	A RAG-based AI chatbot that understands natural language and recommends relevant library books.
Automated Reminders & Fines	High	Students forget due dates; librarians spend too much time calculating fines.	Manual tracking and chasing via phone calls or manual emails.	Automated email and in-app notifications for due dates; automatic daily fine calculation.
Accurate Asset Tracking	High	Books get lost or damaged without clear accountability.	Spreadsheets and paper condition notes.	Strict handover/return workflows requiring librarians to log copy conditions; integrated Write-Off Slips.

3.5 Alternatives and Competition

- **Manual Workflows (Spreadsheets & Paper Cards):** Still used by many universities. Highly prone to human error, lacks real-time updates, and offers no self-service discovery for students.
- **Legacy Library Systems:** Outdated desktop-bound software that lacks mobile responsiveness, membership revenue models, and modern AI recommendation capabilities.